

LAO CAI, Viet Nam

WMO: 488030

Lat: 22.4969N

Lon: 103.9660E

Elev: 318

StdP: 14.53

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.1	52.3	44.4	43.7	56.2	46.9	48.0	55.5	15.0	63.7	12.7	65.6	0.7	140	0.268

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.1	96.8	81.0	94.9	80.9	93.2	80.5	83.3	91.4	82.5	90.7	81.8	89.6	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.4	164.9	87.3	80.6	160.8	86.4	80.0	157.3	85.8	47.8	91.8	46.9	91.0	46.2	90.2	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	9.9	8.1	DB	46.6	101.0	2.9	2.1	44.5	102.5	42.8	103.7	41.2	104.9	39.1	106.4	
			WB	45.6	85.1	2.7	1.6	43.7	86.3	42.1	87.3	40.6	88.2	38.6	89.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.0	63.1	66.3	71.5	78.0	82.4	84.8	84.6	84.0	82.1	77.9	71.7	65.0
	DBStd	9.10	5.90	7.15	6.11	5.42	4.61	3.43	3.14	3.08	3.12	3.79	4.71	4.84
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	259	111	67	15	0	0	0	0	0	0	0	6	60
	CDD50	9493	406	456	668	841	1004	1044	1074	1054	964	864	652	466
	CDD65	4277	51	103	218	391	539	594	609	589	514	399	208	62
	CDH74	41849	116	471	1154	3350	5967	7330	7351	6781	5315	2940	922	152
	CDH80	16675	5	99	292	1200	2631	3383	3229	2861	1999	799	168	9
Wind	WSAvg	2.1	1.9	2.6	2.8	2.9	2.5	1.9	2.0	1.8	1.8	1.6	1.6	1.6
Precipitation	PrecAvg	80.4	1.7	1.8	3.1	5.9	9.8	11.9	15.7	14.9	7.1	4.1	2.5	1.7
	PrecMax	99.0	4.3	5.0	9.1	10.7	18.8	19.9	21.2	25.3	14.4	7.6	6.3	7.3
	PrecMin	63.3	0.0	0.5	0.2	1.2	2.1	6.9	9.6	5.0	3.1	0.0	0.0	0.1
	PrecStd	9.1	1.1	1.2	2.0	2.0	4.4	3.0	2.7	5.3	2.6	2.3	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	86.4	90.0	96.0	99.7	99.5	98.5	97.1	94.4	91.1	87.0	81.1
		MCWB	69.3	71.4	74.2	78.4	80.3	81.8	81.0	81.3	80.0	76.4	74.6	70.5
	2%	DB	77.4	83.3	86.1	91.8	95.8	96.0	95.5	94.9	92.5	88.7	84.3	78.1
		MCWB	67.6	71.0	74.2	78.6	80.8	81.4	81.3	80.7	79.2	76.4	73.3	68.7
	5%	DB	74.2	79.9	83.3	89.1	93.0	93.8	93.6	93.0	90.9	86.9	81.8	75.2
		MCWB	66.4	70.0	73.3	77.8	80.1	81.2	81.2	80.6	78.8	75.9	72.1	67.3
	10%	DB	71.0	76.4	80.2	86.6	90.6	91.8	91.6	91.0	89.0	84.6	79.3	72.2
		MCWB	65.4	69.2	72.4	77.0	79.1	80.9	81.0	80.4	78.4	75.4	71.1	66.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	73.7	76.9	81.6	83.1	84.2	84.6	83.7	82.6	79.9	77.1	72.7
		MCDB	76.3	81.2	84.5	89.9	93.3	92.7	92.0	91.6	89.7	86.2	82.6	78.3
	2%	WB	69.0	72.2	75.4	80.0	81.9	82.9	83.2	82.7	81.5	78.6	75.5	70.6
		MCDB	74.2	79.3	82.7	87.9	91.4	91.2	90.9	90.5	88.3	84.4	80.7	75.2
	5%	WB	67.5	71.1	74.3	78.9	81.2	82.3	82.3	81.8	80.7	77.8	74.3	68.6
		MCDB	72.3	77.7	81.0	86.3	90.0	90.3	90.2	89.3	87.3	83.2	79.2	73.7
	10%	WB	66.1	69.7	73.1	78.0	80.4	81.6	81.6	81.1	79.9	76.9	72.9	66.9
		MCDB	70.8	75.9	79.1	84.9	88.4	89.1	88.9	88.2	86.1	82.1	77.5	71.6
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.7	11.1	12.2	12.9	12.1	11.7	12.2	12.2	10.9	11.9	10.9
		MCDBR	14.5	17.1	15.5	16.3	16.8	15.6	15.3	15.6	15.2	14.3	15.0	14.7
		MCWBR	5.7	6.1	5.3	5.5	4.9	4.4	4.4	4.7	4.6	4.3	5.5	6.2
	5% WB	MCDBR	12.9	14.5	14.0	14.5	14.7	13.4	13.6	13.9	12.9	11.6	11.9	12.8
		MCWBR	5.9	6.1	5.4	5.4	5.0	4.6	4.8	5.0	4.5	4.3	5.4	6.4
Clear-Sky Solar Irradiance	taub		0.576	0.620	0.851	0.801	0.547	0.507	0.500	0.563	0.590	0.659	0.580	0.613
	taud		1.780	1.717	1.347	1.472	2.053	2.187	2.195	1.994	1.895	1.696	1.818	1.703
	Ebn at Noon		216	217	177	191	245	253	254	239	230	205	213	199
	Edh at Noon		64	72	109	98	54	47	47	57	62	73	61	67
All-Sky Solar Radiation	RadAvg		795	1067	1202	1450	1505	1361	1397	1420	1338	1060	1036	809
	RadStd		134	199	111	159	162	126	96	108	85	104	106	95

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Station Only	+0.63	N/A	-1.55	+1.13	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.72	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	+347	+265

Nomenclature: See separate page